

Pathology Examination

Third Lecture: Chronic Inflammation & Tuberculosis (TB)

Part I: Multiple Choice Questions (MCQs)

1. A granuloma is a type of chronic specific inflammation characterized by focal accumulation of:

- A) Neutrophils and fibrin
 - B) Basophils and mast cells
 - ✓ C) Modified macrophages (epithelioid cells), lymphocytes, giant cells, and fibroblasts
 - D) Red blood cells and platelets
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2. Grossly, tuberculous granulomas appear as:

- A) Large black masses
 - ✓ B) Scattered tan-yellow granulomas
 - C) Small blue vesicles
 - D) Diffuse red streaks
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3. Microscopically, the center of a tuberculous granuloma shows caseation necrosis in the form of:

- A) Fluid-filled cavities
 - ✓ B) Cheesy grayish-yellow, structureless homogeneous eosinophilic material
 - C) Aggregates of neutrophils
 - D) Calcified bone
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4. The most accurate method for diagnosing tuberculosis (TB) is:

- A) Tuberculin skin test
 - B) History of contact
 - C) Tissue biopsy alone
 - ✓ D) Detection of TB bacilli by sputum examination, culture, or PCR
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Part II: Essay & Short Answer Questions

Q1. Discuss the Etiology of Granuloma

Answer:

The causes of granuloma are classified into:

1. Infectious Causes:

- **Bacterial:** Tuberculosis, Leprosy
- **Parasitic:** Bilharziasis
- **Fungal:** Histoplasmosis

2. Non-Infectious Causes:

- **Foreign body:** Silicosis, surgical sutures
- **Autoimmune diseases:** Sarcoidosis, Crohn's disease

Q2. Compare between Primary and Secondary Tuberculosis (TB)

Feature	Primary TB	Secondary TB
Exposure	Unsensitized (usually children)	Previously sensitized (usually adults)
Cause	First infection	Re-infection or reactivation
Organs	Lung, intestine, tonsil, skin	Any organ
Lymphadenitis	Prominent	Minimal
Tissue destruction	Mild	Marked
Tissue reaction	Slow	Accelerated (hypersensitivity)
